Data availability

Chemical

pH (1:5)
Macronutrients (N,Ca, P...)
Micronutrients (Fe,Mb,Cu...)

Biological

Organic Matter (%)
C/N Ratio
Organic carbon (%)

Physical

Clays (%)
Silts (%)
Sands (%)
Texture

Risks

Working on sandy, wind-prone soils makes permanence a continuous challenge. The programme therefore applies conservative baselines, uses a buffer to absorb unexpected losses, and adjusts practice rules in high-risk sub-zones (wind, floods, salinity). Financially, it mitigates volatility by stacking carbon credits with value-chain premia and multi-year offtake agreements, while administrative risks are reduced through single-window processes and standardised digital templates. Advisory support, resilient cover mixes and contingency plans ensure farmers stay engaged even under drought or saline stress.



Preliminary results

Early outcomes point to incremental but lasting improvements: higher soil organic carbon, stronger aggregation and infiltration, and reduced erosion, translating into more resilient yields. Economically, farms secure a predictable public support floor complemented by new revenue from credits, while confidence grows as MRV costs fall with stable sampling designs and remote-sensing integration. Institutionally, the model demonstrates how blending action-based and results-based instruments within a district framework can deliver credible certification and practical benefits on challenging sandy soils.

Economic perspective

The Sustainability District treats carbon credits as a component, not the whole story. SOM increases and reduced erosion are translated into carbon removal/emission reduction claims that can be certified and marketed; meanwhile, the same practice set underpins a product-quality and stewardship narrative that coastal buyers can recognise (labels, standards, supplier programmes). This creates two revenue streams:

1. Result-based income from verified carbon outcomes; and
2. Value-chain premia (or preferential procurement) for produce carrying the District's stewardship claim.

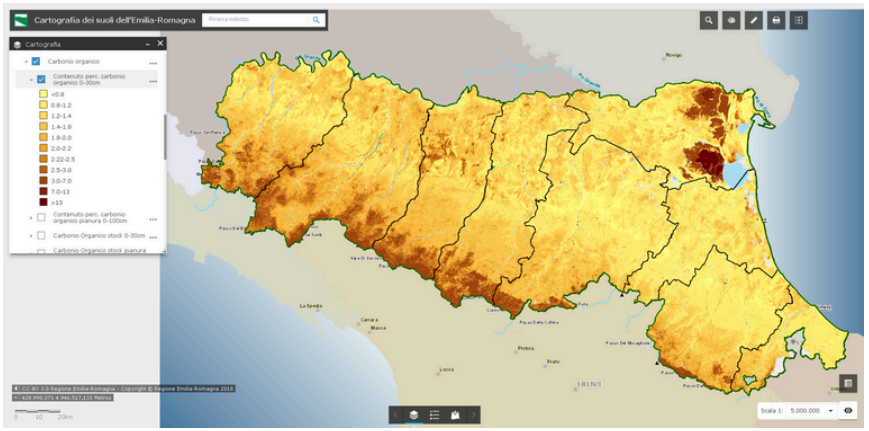
Importantly, the case embraces hybrid incentives. A public, action-based floor (eco-schemes/AES for cover crops, reduced tillage, landscape features) provides predictability and helps cover extra costs. On top, the result-based carbon component

Policy and governance

The Italian case highlights farmers' reluctance towards long, volatile carbon contracts and complex monitoring, showing preference for short-to-medium commitments with renewal options and per-hectare payments that reflect real costs. Policy instruments should therefore combine eco-scheme or AES-based support with seed funding to cover initial MRV and protocol design, de-risking early participation.

Governance is structured around a District Operator (co-operative, consortium, producer organisation) acting as aggregator: it standardises practices, manages data and sampling, negotiates with buyers/certifiers, operates risk buffers and distributes payments. Regional authorities and paying agencies provide action-based floors, oversee compliance, and co-finance MRV. Certification bodies validate quantification and permanence, while water and land authorities address drainage and salinity risks. Research and advisory partners contribute protocols, decision-support tools and QA/QC.

District of the Sands



novasoil

The NOVASOIL project aims to show the benefits for society and the environment of investing in soil health.

The main output of the project will be a tool to analyse the suitability of different business cases that promote soil health.

This tool will be based on best practice examples in Europe and other countries, as well as on the needs and demands of society.

A model for multifunctional and sustainable development of marginal areas

Summarize



Along the Emilia–Romagna coast, farming takes place on highly sandy soils that are productive yet fragile. The area faces chronic pressures: salinisation driven by sea-water intrusion and drought, subsidence linked to hydrology and land use, gradual loss of soil organic matter (SOM) and, in places, legacy contamination. The case study sets an explicit dual ambition. Agronomically, it aims to rebuild organic matter and water-holding capacity so that fields weather heat and irregular rainfall without sacrificing yield stability.

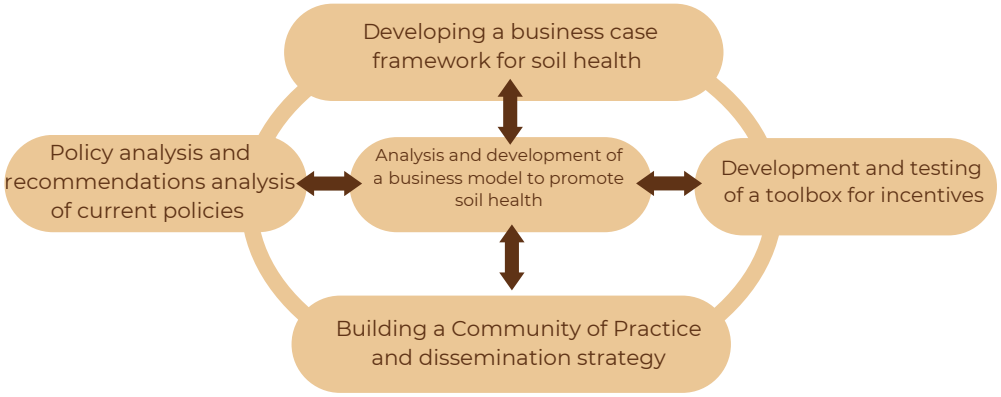


Figure 1. Structure of the NOVASOIL project

Objectives

In the NOVASOIL Project, we will focus on learning from experiences in the project by exploring the effectiveness of the actions made in terms of soil fertility and health improvement, productivity rising, resilience and landscape valorisation. We will investigate the possibility to develop a business models aiming at adding value to the agronomic management aimed at soil health conservation with innovative crop products through certification and production standards.

Case studies properties

Potential inhibitors

Competition with products from other countries with a lower cost,
Market changes
Extreme weather conditions (extreme rainfall, floods, and drought)

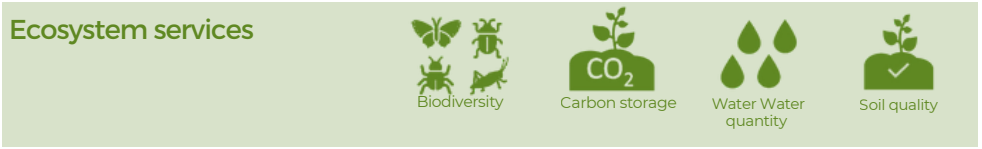
Potential enablers

Local, organic certified organic and fully traceable final products
Social awareness interested in high quality and healthy product
Existing farmer network
Higher revenue for farmers compared to previous cropping system

Potential stakeholders

Emilia-Romagna Region - Regional administration | Private companies | Farmers |Farmer association | Accademia dei Georgofili





Specific objectives

SO1 Multi-dimensional portfolio development of business cases and good examples for investing in soil health in Europe and beyond.

SO2 Analysis of current models around the world for creating new incentives and new revenues from healthy soils.

SO3 Apply a co-development approach to modelling that promotes soil health.

SO4 Develop a prototype incentive toolbox and test network.

SO5 Analysis of current policies and technologies related to soil health and income provision for farmers.

SO6 Developing a community of practice around investors and key soil health actors.

SO7 Increase interest in soil health through a digital marketing strategy plan

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INNOVATIVE BUSINESS MODELS FOR SOIL HEALTH

https://novasoil-project.eu

